MUHAMMAD ASAD SARFRAZ

AI Engineer | LLMs, RAG Systems & Intelligent Automation

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PROFESSIONAL SUMMARY

AI Engineer and senior backend engineer with 7+ years of experience building production-grade systems and 2+ years of hands-on LLM integration and AI pipeline development. Experienced in designing and deploying RAG systems, LLM-powered applications, and agentic workflows using LangChain, LangGraph, OpenAI, and Anthropic APIs. Strong Python engineering foundation with deep expertise in REST APIs, vector-adjacent search (Elasticsearch), containerization (Docker/Kubernetes), and cloud deployments on AWS. Previously contributed to foundational LLM improvement projects for FAAMG partners at Turing, working on model evaluation, prompt engineering, and AI dataset creation.

TECHNICAL SKILLS

AI & LLM: LangChain, LangGraph, OpenAI API, Anthropic Claude API, RAG pipeline design, prompt engineering, model evaluation, AI agents, function-calling, ChatGPT Plugins

Vector & Semantic Search: Elasticsearch (semantic + keyword search), embeddings, vector search concepts, semantic ranking

Python: FastAPI, Django, Flask — production-grade backend systems, async APIs, data pipelines

Workflow Automation: n8n, LangGraph agent orchestration, Sidekiq, multi-step API workflow design, agentic tool-use

Backend & APIs: REST API design, microservices, SOA, Ruby on Rails, PostgreSQL, Redis, MySQL

DevOps & Cloud: Docker, Kubernetes, AWS, Heroku, CI/CD pipelines, Linux/Unix, Bash

Testing & Quality: TDD, unit & integration testing, automated test suites, LLM output evaluation frameworks

Other: TypeScript/React, Git, Jira, technical documentation, API specifications

PROFESSIONAL EXPERIENCE

LLM Trainer — Python & Ruby | Turing (FAAMG Partner Projects) *Nov 2024 – Jan 2026*

- Designed and evaluated LLM prompts, multi-turn conversations, and agent function-calling scenarios for foundational model companies — directly contributing to model fine-tuning and benchmarking datasets.
- Built prompt engineering and validation frameworks to evaluate model outputs across reasoning, coding, and tool-use dimensions, identifying failure modes and edge cases systematically.
- Collaborated with FAAMG engineering teams on AI-enabled delivery solutions, applying knowledge of OpenAI and Anthropic model behavior to improve instruction-following reliability.
- Developed automated QA pipelines for AI training data quality control, supporting continuous improvement of LLM capabilities.

Backend Lead / Senior AI-Integrated Engineer | DevDen *2023 – Present*

- Architected and deployed AI-powered backend systems integrating LLM APIs (OpenAI, Anthropic) into production FastAPI and Rails services, including automated content generation and intelligent search ranking.
- Built RAG pipeline foundations using Elasticsearch as the retrieval layer with semantic search capabilities — enabling context-aware, relevance-ranked query results over large property datasets.
- Designed LangChain-based agent workflows for automated data processing and multi-step API orchestration, replacing manual workflows with intelligent automation.

- Led integration of n8n automation workflows connecting internal systems with external APIs — significantly reducing manual operational overhead.
- Owned CI/CD pipeline design and containerized deployments via Docker and Kubernetes on AWS, ensuring reliable AI service delivery in production.
- Drove performance optimization: response times reduced from 60s → 8.3s and page payloads from 46 MB → 3.6 MB — demonstrating systematic AI system performance tuning.
- Led and mentored a team of 5 engineers, setting architectural direction and enforcing quality standards across AI and backend systems.

Software Engineer — Full Stack | DevDen *2021 – 2023*

- Integrated third-party APIs and external services (Google APIs, Stripe, channel managers) into Python and Rails backends — building the REST API and integration experience foundational to LLM tool-use development.
- Built microservices architecture enabling modular, independently scalable backend components — directly applicable to AI service decomposition patterns.
- Developed automated multi-step business workflows requiring complex conditional logic, state management, and failure handling — the same design patterns used in agentic AI systems.
- Implemented Elasticsearch-powered search with custom ranking and filtering — hands-on experience with the retrieval layer central to RAG architectures.

KEY PROJECTS

LLM Agent & RAG Pipeline Development *(LangChain, LangGraph, OpenAI, Anthropic, FastAPI)*

- Designed multi-step LLM agent workflows using LangChain and LangGraph for automated reasoning and task execution across production systems.
- Built RAG pipelines combining Elasticsearch retrieval with LLM generation, enabling context-grounded, accurate AI responses over proprietary datasets.
- Implemented prompt engineering frameworks for consistent, reliable LLM outputs with structured evaluation and quality scoring.

Nice2Stay — AI-Enhanced Global Booking Platform *(Python/Flask, Elasticsearch, LLM integrations, AWS)*

- Integrated intelligent search ranking using Elasticsearch semantic capabilities, combining vector-adjacent scoring with traditional filters for property discovery.
- Deployed containerized AI services on AWS with high availability, managing AI system reliability and performance monitoring in production.

n8n Workflow Automation Platform *(n8n, REST APIs, Python, LLM integrations)*

- Built complex n8n automation workflows integrating LLM APIs, internal databases, and external services — end-to-end AI-powered process automation.
- Designed trigger-based pipelines with conditional branching, error handling, and retry logic for reliable automated execution.

EDUCATION

B.Sc. Computer Science | National University of Computer & Emerging Sciences (FAST) *2017 – 2021*

- Dean's Honor List: Fall 2018, Spring 2018, Fall 2020, Spring 2021

Cloud Native & Mobile Web Computing | Presidential Initiative for AI & Computing (PIAIC) *2019 – 2021*

- Cloud-native architecture, containerization, API design, and modern AI application paradigms.

ACHIEVEMENTS

- Bronze Medal — NUCES CFD Campus, Fall 2020
- Runner Up — Nascon'19 Battle of Brains, NUCES ISB Campus
- Code Debug Winner — Kath'18, University of Gujrat
- Speed Programming Competition — IRIS 2.0, RIPHA International